

From Data to Decision

Session 2 · How an AI system is actually built

Capabilities & limits

Anatomy of a system

Building the model

Ten scenarios

Agenda

- I Recap — where Session 1 left off
- II What AI can and cannot do
- III The anatomy of an AI system — task, modality, paradigm, model class
- IV Building the fraud model, stage by stage**
- V The same anatomy, across industries
- VI Your turn — ten scenarios, three calls each
- VII Wrap-up and hands-on work

Tonight's throughline: every capability AI has — and every limit — traces back to its data.

Where we are — the three steps from Session 1

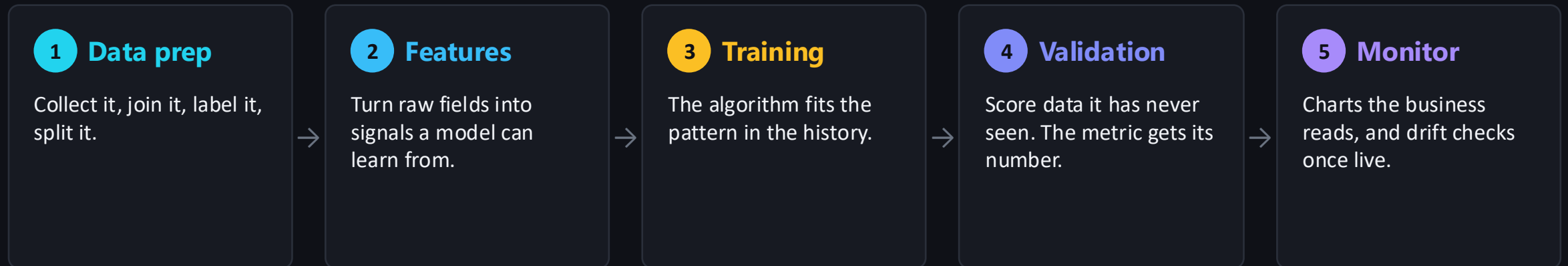
- 1 Map the business process, and find the decisions in it**
Inputs → Process → Decision → Action → Outcome. Six decisions inside one online order.
- 2 Judge whether AI is a good fit for each decision**
The test: would it behave differently with another year of data? Four of our six were eliminated.
- 3 Design the AI application for the decisions that fit**
Data → Model → Metric → Human. The payment-screening decision, end to end.

Tonight deepens step 2 and step 3

Judging whether AI fits requires knowing what AI can actually do. Choosing a model requires knowing how machines learn. That is tonight.

Tonight: how the model actually gets built

Last week you decided the payment-screening decision was worth building. Tonight we build it — stage by stage, on the same online order.



Most of the evening is spent in the first two stages — and so is most of a real project

Practitioners consistently report that collecting, cleaning and engineering the data takes the bulk of the work. Training, the stage everyone pictures, is often the shortest.

Same order, same decision, same fraud analysts. Only the altitude changes.

SECTION II

Definition and evolution of AI

What AI is, where it came from, and — as a consequence — what it can and cannot do.

What artificial intelligence is — a working definition

AI is the ability of a machine to perform tasks that normally require human intelligence — learning, reasoning, problem-solving, perception, and understanding language.

Artificial Intelligence

The whole ambition — machines doing tasks that need intelligence. Includes rules-based systems.

Machine Learning

The subset that learns its behaviour from data instead of following written rules.

Deep Learning

The subset of ML using many-layered neural networks — images, speech, language.

Nested, not synonyms: all deep learning is ML, all ML is AI — and none of the arrows reverse.

Seventy years, two winters, one spring

1956–1970s

Birth and first winter

Named at the Dartmouth Workshop, 1956. Early programs reasoned with logic. Promises outran hardware — funding collapsed.

1980s–1990s

Expert systems, second winter

Hand-written rules captured expert judgment — until the rules could not keep up with the world. Collapsed again.

2000s–2010s

The machine learning spring

Big data + GPUs + better algorithms. Systems stopped being told rules and started learning them.

2017–today

Transformers and GenAI

One architecture, trained on the internet's text, made language a machine capability.

Compute Unlocked Modern AI

Many AI ideas existed long before the hardware could run them at useful scale. GPUs, faster chips, cloud computing, and cheaper storage, combined with more data and better algorithms, turned those ideas into practical systems.

What AI genuinely does well

- 1 Finds patterns too subtle or too numerous for people**
A fraud signal spread across forty features, or a shadow on one X-ray in ten thousand.
- 2 Applies a judgment consistently, at any volume**
The ten-thousandth payment is screened exactly like the first. No fatigue, no Friday afternoon.
- 3 Answers in milliseconds, around the clock**
Decisions that must happen while a customer waits — or forty thousand times a month.
- 4 Produces fluent drafts of text, images, and code**
Generation: a starting point in seconds, for a human to shape and verify.

All four are the same trick: a pattern learned from data, applied fast and without tiring.

What AI cannot do — and why

IT CANNOT	BECAUSE	SO KEEP THE HUMAN WHEN
Handle what its data never showed	It interpolates from history. A truly novel situation has no pattern to match.	Situations are new or rare
Know why — only what correlates	It learns associations, not causes.	The why is the question
Weigh values or take responsibility	There is no number for “is this fair to this customer?” Someone must own the call.	Stakes are ethical or legal
Know when it is wrong	It outputs confidence, not truth. A wrong answer arrives as fluently as a right one.	Errors are costly to miss
Fix bad data	Bias in the history becomes bias in the model — faithfully reproduced at scale.	History itself is suspect

Traditional ML, Generative AI, and AGI

The easiest distinction is the output: prediction, new content, or hypothetical general capability.

Traditional machine learning

Predicts a defined output from historical examples.

Returns a label, number, ranking, or forecast.

Example: score a payment for fraud risk.

STATUS · Deployed today

Generative AI

Creates new content from a prompt and context.

Returns text, images, audio, video, or code.

Example: draft a customer reply.

STATUS · Deployed today

Artificial general intelligence

Would learn, reason, and adapt across many domains with human-level flexibility.

No verified business system has this capability.

STATUS · Hypothetical

SECTION III · 14 min

The anatomy of an AI system

Four dimensions describe any system you will meet: the task, the data going in, how it learned, and what was built.

Any AI system is four choices, made independently

Four dimensions. One answer on each describes a system completely — and each varies independently of the others.

TASK

what shape comes out

Classification

Regression

Forecasting

Anomaly detection

Generation

DATA MODALITY

what goes in

Tabular

Text

Image / video

Audio

Time series

LEARNING PARADIGM

how it learned

Supervised

Unsupervised

Self-supervised

Reinforcement

MODEL CLASS

what gets built

Linear / tree

Gradient boosting

CNN

Transformer

Solver

The outlined row is our fraud model — and each row moves on its own

Classification, on tabular data, learned supervised, built as gradient boosting. Change only the modality to images: the task is still classification, but the model becomes a CNN and somebody must label ten thousand photographs.

Task — the shape of what comes out

Nine tasks cover essentially all deployed AI. Last week's four patterns are the business-level grouping over these.

TASK	WHAT COMES OUT	AN EXAMPLE
Classification	One of N labels	Approve, review, or decline this payment
Regression	A number	How many days will this delivery take?
Forecasting	A number over time	Units of this item sold next quarter
Anomaly detection	“Is this one unusual?”	A sign-in from an impossible location
Ranking	An ordered list	Which products to show this customer
Clustering	Groups nobody defined	Customer segments nobody wrote down
Extraction	Fields out of a document	The order number inside a customer email
Generation	New text, image, or code	Draft the reply to that customer
Optimization	An assignment or schedule	Which warehouse, which carrier

Modality — the kind of data going in

The input format drives the model choice, the preparation work, and much of the project cost.

Tabular STRUCTURED TYPICAL MODEL Gradient boosting PROJECT REALITY Lowest cost, fastest to train, and easiest to explain.	Text LANGUAGE TYPICAL MODEL Transformer / LLM PROJECT REALITY Needs trusted source documents and retrieval to reduce invented answers.	Image / video VISUAL TYPICAL MODEL CNN PROJECT REALITY Requires thousands of correctly labeled images.
Audio / speech SPOKEN TYPICAL MODEL Speech-to-text + text model PROJECT REALITY Transcription errors carry into every later step.	Time series TEMPORAL TYPICAL MODEL Forecasting model PROJECT REALITY Seasonality and drift make old patterns expire.	Graph CONNECTED TYPICAL MODEL Graph model PROJECT REALITY Specialist modeling for fraud rings and supply networks.

● WHY MODALITY MATTERS
Model choice · data preparation · project cost

● OUR FRAUD MODEL
Tabular data → fast training and explainable drivers

Machine learning paradigms — how models learn from data

Supervised

Answered examples. Thousands of inputs paired with the right answer. The model learns input → answer. Powers classification and prediction — the bulk of business AI.

Unsupervised

Find structure. No labels at all — the model groups what is similar or flags what is odd. Customer segments, anomaly detection.

Reinforcement

Trial and reward. An agent acts, gets a score, adjusts. How game-playing systems and some robotics learn. Rare in week-to-week business analysis.

The one question this answers for you as an analyst

“Do we have labels?” Supervised learning — most of what you will propose — needs history where the right answer is already recorded. That is why Session 1's Data step asked whether the data is labelled.

Four network families, four kinds of data

1 CNN

Convolutional network

Sees images. Scans for local features — edges, textures, shapes. X-rays, defect photos, faces.

2 RNN

Recurrent network

Reads sequences in order. Time series, sensor streams. Largely succeeded by transformers for text.

3 GAN

Generative adversarial

Two networks — one forges, one detects — train each other. Synthetic images and data augmentation.

4 Transformer

Attention-based

Weighs every word against every other. The architecture behind BERT, GPT, and modern GenAI.

Putting it together — pattern, paradigm, model, data

PATTERN	USUALLY LEARNED BY	TYPICAL MODEL	DATA IT NEEDS
Classification	Supervised	Gradient boosting · CNN for images	Labelled examples — with correct labels
Prediction	Supervised	Gradient boosting · time-series models	History with actual outcomes recorded
Optimization	Often not learned at all	Solver (linear / constraint programming)	A precise objective and constraints
Generation	Self-supervised at scale	LLM (transformer), grounded	Vast text + your documents for grounding

SECTION IV · 65 min

How the fraud model gets built

Data prep → features → training → validation → monitoring. All five stages, on the payment decision from last week.

Data ingestion — where the data actually comes from

1 Order database

Items, amounts, addresses, timestamps, the customer's account history. Ours, and reliable.

2 Payment gateway

Card metadata, authorization results, device fingerprint. A third party's system, on their API.

3 Chargeback file

Which orders were later disputed and lost. Arrives monthly, as a spreadsheet, from the bank.

The join is the work, and it is where the first surprises appear

Three systems, three different ideas of what identifies an order. Payment records key on a transaction ID; chargebacks arrive with a bank reference and a date. Matching them is a week of somebody's life, and the match is never 100%.

The label problem — what is the truth here?

What we have

Every decision an analyst made. Approved, reviewed, declined — with a timestamp and a name.

What we need

Whether the order was actually fraudulent. That is a different fact, and it lives somewhere else.

Train on decisions and you rebuild the analysts — including their mistakes

If the label is “what the analyst decided,” the model learns to imitate the team. Every bias, every Friday-afternoon miss, reproduced at forty thousand orders a month and defended as objective. You need the outcome, not the record of the decision.

THE ONLY HONEST LABEL

AND ITS PRICE

A chargeback

The customer disputed it and the bank agreed. This is real fraud, confirmed.

Arrives 60–90 days late

So last quarter's orders are still unlabelled. Your training data is always months behind today.

Misses what we blocked

Orders we declined never got a chance to be fraud. We will never know if we were right.

Splitting the data

1 Training

Months 1–30. The model reads these and fits the pattern.

2 Validation

Months 31–33. Used to tune settings while building.

3 Test

Months 34–36. Touched once, at the end. The honest score.

Split by time, not at random

A random split lets the model train on March and be tested on February — learning from the future to predict the past. Fraud tactics move; a model that has seen next month's tricks will look far better in testing than it will ever be in production. Hold out the most recent months, always.

Feature engineering — turning records into signals

A model cannot use “an order.” It uses numbers you compute from the order. Those are features, and choosing them is the craft.

THE FEATURE

WHAT IT IS ACTUALLY MEASURING

Amount vs. this customer's usual

Not “is \$840 big?” — is \$840 big for this person? A ratio, not a number.

Billing / shipping mismatch

Same name, different state. Legitimate sometimes; a strong signal in combination.

Account age at order time

Fifteen minutes old is not the same as fifteen months old.

Device seen before?

Has this browser and machine ever placed a successful order for us?

Orders in the last hour

Velocity. One order is a purchase; nine in ten minutes is a card being tested.

Hour of day, local time

Weak alone. Useful combined with everything above.

How domain knowledge becomes model-ready data

A model cannot use intuition directly. The team must translate an expert observation into measurable features.

1 Fraud analyst observes

“Three failed card attempts followed by a successful payment may indicate card testing.”

Experience identifies a behavior that may be worth measuring.



2 Data scientist creates

From payment timestamps and outcomes:

failed_attempts_10m = 3
success_after_failures = yes

The same calculation runs for every order.



3 Model learns

Question tested against labeled history:

Does this pattern predict confirmed fraud? If so, the model learns how strongly it should influence the risk score.

One signal, not a verdict

Three failed cards do not automatically mean fraud. The model weighs this pattern with amount, device history, account age, and other evidence.

A model cannot learn from expertise until that expertise is translated into measurable data.

Data leakage — the feature that is not there when you need it

A worked example: the feature that is 99% accurate and completely useless

Somebody adds “was this order refunded?” to the training data. The model becomes almost perfect — refunds and fraud go together. Then it goes live, and at the moment of decision no order has been refunded yet. The column is empty. Accuracy collapses from 99% to worse than the old rules.

THE TEST

Would this value exist at 2:14pm on Tuesday, the second the customer clicks Buy?

Was it recorded after the decision, or because of it?

APPLIED TO OUR FRAUD MODEL

Account age — yes. Device history — yes. Chargeback status — no, that is 60 days away.

“Reviewed by analyst” exists only because we flagged it. Feed that in and the model learns to predict our own queue.

Why it survives so long

Leakage does not look like a bug. It looks like success. The validation score is excellent, everyone is pleased, and the problem only appears weeks after deployment when the numbers quietly fail to reproduce.

The same stage, on images and text

Feature engineering is what you do to tables. For images and language, the network learns the features itself — and that single difference explains the last ten years of AI.

1 Tables

A person computes the columns: ratios, counts, flags. Slow, judgment-heavy — and the result is explainable line by line.

2 Images

You cannot hand-craft “looks like a tumour.” A CNN learns edges, then shapes, then structures, from labelled examples.

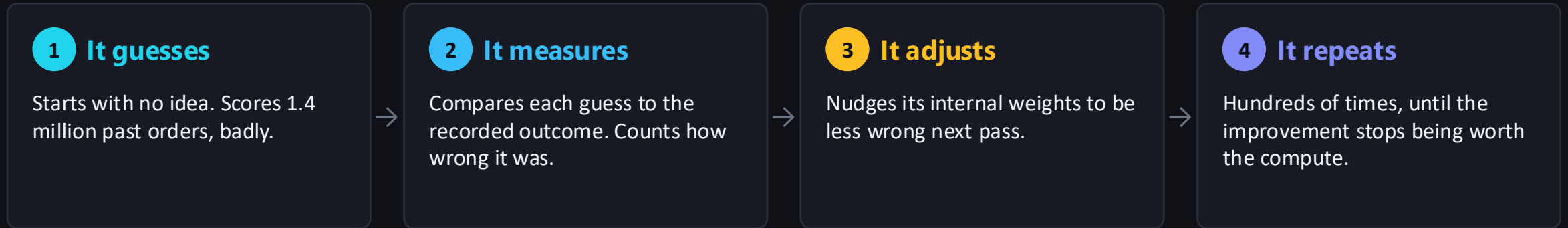
3 Text

Words become vectors that place similar meanings near each other. The model learns the representation; you supply the grounding.

This is why deep learning took over vision and language — and did not take over spreadsheets

Where features are impossible to hand-craft, learning them is transformative. Where a domain expert can name the six columns that matter, gradient boosting on those six is cheaper, faster, more accurate, and defensible in front of a regulator. Our fraud model is firmly in the second camp.

Training — what “fitting a model” actually means



OUR CHOICE

WHY, IN ONE LINE

Gradient boosting

Tabular data with named features — the situation it is best at, by a distance.

Not a neural network

It would need far more data, cost more to run, and could not show a regulator why it declined an order.

~1.4M labelled orders

Three years of history, minus the six months held out for testing.

Validation — score it on the months it has never seen

One held-out test month. 10,000 orders went through. 100 of them were really fraud.

	ACTUALLY FRAUD	ACTUALLY CLEAN	ROW TOTAL
Model flagged it	80	170	250
Model cleared it	20	9,730	9,750
Column total	100	9,900	10,000

Precision = $80 \div 250 = 32\%$

Of everything we flagged, this much was really fraud. The other 68% were good customers we interrupted.

Recall = $80 \div 100 = 80\%$

Of all the fraud that actually happened, this much was caught. Twenty cases went straight through.

The same numbers, read as accuracy

Our model

Right on $80 + 9,730 = 9,810$ of 10,000 orders.

Accuracy: 98.1%

And it catches 80% of the fraud.

A model that says “never fraud”

Right on all 9,900 clean orders. Wrong on all 100 frauds.

Accuracy: 99.0%

And it catches nothing at all.

The do-nothing model scores higher, and is worth nothing

This is class imbalance, and it is the formal name for the trap. When the positive case is rare, accuracy measures the base rate, not the model. Fraud, defects, disease, churn, safety incidents — everywhere the interesting case is a small minority, accuracy is the metric that makes a useless system look excellent.

Visualization and monitoring — the stage that never ends

1 What the team sees

A score for every order, a queue sorted by it, and the features that drove each score. Not a black box on a screen.

2 The threshold

One control, and it is theirs. Slide it down to catch more fraud and review more good customers. Up for the reverse.

3 Concept drift

Fraud tactics move, so the relationship the model learned decays. Precision and recall are re-measured monthly against fresh chargebacks.

The threshold is a business decision wearing technical clothing

Moving it trades false alarms against missed fraud. Nothing in the mathematics says which is worse — that depends on what a blocked customer costs against what a chargeback costs, and only the business can answer it. The data scientist builds the slider. The fraud lead decides where it sits.

A model is not delivered when it is accurate. It is delivered when somebody can run it, and change it.

The whole build, on one page

STAGE	WHAT HAPPENS	WHO IS IN THE ROOM	WHAT GOES WRONG
Data prep	Join orders, payments and chargebacks. Split by time, not at random.	Data engineer + fraud team	Labels record decisions, not outcomes
Features	Compute the signals: amount vs. usual, address mismatch, velocity, device.	Data scientist + fraud analysts	Leakage — a value that is not there yet
Training	Gradient boosting fits ~1.4M labelled orders.	Data scientist	Rarely the problem. Usually the shortest stage.
Validation	Score the held-out months. Precision 32%, recall 80%.	Data scientist, checked by the business	Accuracy on a rare event flatters a useless model
Monitoring	Ranked queue, a threshold the team owns, monthly drift checks.	Everyone — this is the handover	Nobody owns the threshold, so nobody moves it

Four of the five stages need somebody who understands fraud. Only one needs somebody who understands models.

SECTION IV

The same patterns, across six industries

The landscape in three industries — healthcare, finance, manufacturing — each with a use case walked end to end.

EVERY EXAMPLE TONIGHT IS CLICKABLE AT HOME

First, the map — six workflows in Applied AI Studio

WORKFLOW	INDUSTRY	THE DECISION INSIDE	PATTERN
Online Order	Retail	Is this payment fraudulent?	Classification
Maintenance Triage	Manufacturing	Which failure report is urgent?	Classification
Customer Identity Review	Finance	Is this applicant who they claim?	Classification + human
Resident Service Intake	Government	Which department, how urgent?	Classification
Fleet Routing	Transportation	Which vehicle takes which stop?	Optimization
Referral-to-Appointment	Healthcare	Who needs scheduling first?	Prediction + human

Open any of these in Applied AI Studio — each shows the workflow map, the AI-fit score, and the design blueprint.

Different industries, same anatomy: a process, a decision inside it, and a pattern that fits — or does not.

AI in healthcare — where it is applied

Rising costs, aging populations, scarce specialists — and workflows full of decisions.

APPLICATION	WHAT IT DOES	PATTERN
Medical image analysis	Reads X-rays, CTs, retinal photos for signs of disease	Classification · CNN
Predictive analytics	Flags readmission risk and deterioration before it happens	Prediction
Drug discovery	Screens millions of candidate molecules for promising ones	Prediction + generation
Personalized medicine	Matches treatment to the patient's history and genetics	Prediction
Robotic surgery	Steadies and assists the surgeon's instruments	Vision + robotics

Every row keeps a clinician in the loop — in this industry the human is a design requirement, often a legal one.

Use case — diabetic retinopathy screening

Problem

A leading cause of preventable blindness. Early detection saves sight — but screening needs specialists, and many clinics have none.

Solution

IDx-DR (now LumineticsCore): autonomous screening at the primary-care visit. In 2018, the first AI the FDA authorized to decide without a physician reading the image.

Data

Large sets of retinal photographs, graded by ophthalmologists. Labelled — so supervised learning.

Model

A convolutional neural network — the data is images, so deep learning earns its cost.

Benefits

Screening where no specialist works · earlier detection · specialist time redirected to treatment.

Challenges

Bias if training images do not match the patient population · a 900-patient trial to clear the FDA · out-of-scope images must be referred, never guessed.

FDA De Novo authorization, April 2018 — the first autonomous AI diagnostic in any field of medicine.

AI in finance — where it is applied

An industry made of data, decisions, and regulators reading over your shoulder.

APPLICATION	WHAT IT DOES	PATTERN
Fraud detection	Scores every transaction, in real time, for fraud risk	Classification
Credit risk	Estimates the likelihood a loan will be repaid	Prediction
Algorithmic trading	Executes strategies at machine speed	Prediction + optimization
Customer service	Answers routine questions, drafts responses	Generation + human
Personalized advice	Tailors financial guidance to the customer	Prediction + generation

Credit risk carries a warning: models must be explainable and provably fair — Fair Lending law reaches the algorithm. Module 4's whole story.

Use case — real-time card fraud detection

Problem

Card fraud costs billions, and every false alarm blocks a real customer's purchase. Manual review cannot touch real-time volumes.

Solution

A supervised classifier scores every transaction in milliseconds: approve, challenge, or decline — with analysts reviewing the ambiguous band.

Data

Years of transactions — amount, merchant, location, time, device — labelled by outcome: chargebacks mark the fraud we missed.

Model

Gradient boosting or random forest on tabular features. Deliberately not deep learning: tables, and regulators ask why.

Benefits

Lower fraud losses · fewer good customers blocked · instant decisions at any volume.

Challenges

Fraud is rare, so labels are imbalanced — the accuracy trap · patterns move, so the model drifts · every decline may need explaining.

This is Session 1's payment decision at industrial scale. You have already designed this system — say so in interviews.

SECTION VI

Your turn — ten scenarios, three calls each

For each: (1) Is AI appropriate at all? (2) If yes, which pattern? (3) What data would it require?

Scenarios 1–5

- 1 Every morning, assign 40 delivery vans to 600 stops**
Routes must respect driver hours, vehicle capacity, and delivery windows.
- 2 Flag suspicious sign-ins to the company's systems**
Odd hours, new devices, impossible travel — thousands of sign-ins a day.
- 3 Reject order forms that are missing required fields**
Name, address, payment — the required list is written in the procedure manual.
- 4 Predict which patients will miss next week's appointments**
The clinic loses 18% of slots to no-shows and wants to overbook intelligently.
- 5 Draft first replies to routine customer emails**
Password resets, delivery questions, return policies — 900 emails a day.

Work in pairs. Three calls per scenario — and “not AI” is a full answer if you can say why.

Scenarios 6–10

- 6** **Decide which suppliers' invoices to pay first under contract terms**
Payment order follows negotiated terms, discounts, and due dates.
- 7** **Forecast demand for each store item next quarter**
8,000 SKUs, seasonal swings, promotions, and a new competitor across the street.
- 8** **Route incoming job applications to the right hiring team**
Eleven teams, high volume — and last year's routing lived in one coordinator's head.
- 9** **Read defects off the production line from camera images**
A scratch or misprint every few thousand units, at line speed.
- 10** **Decide whether to grant a goodwill refund to a furious longtime customer**
Policy says no; the relationship says maybe; the amount is \$840.

Full worked answers for all ten are in this week's handout — argue with them.

Key takeaways

- 1 Every capability and every limit traces back to the data**
Pattern-finding, consistency, speed, fluency — and brittleness, bias, drift, confident error. One source for all of it.
- 2 Four dimensions describe any system**
Task, data modality, learning paradigm, model class. They move independently — change the modality and the model class changes with it. Ask for all four.
- 3 Most of a project is data and features, not training**
Four of the five build stages need somebody who understands the business. Exactly one needs somebody who understands models.
- 4 The label is the first thing to interrogate**
We know what analysts decided, not what was true. Train on decisions and you rebuild the team, mistakes included.
- 5 Accuracy lies when the thing you want is rare**
“Never fraud” scores 99%. Ask for precision and recall — and know that the threshold between them is a business decision.